

BENCHMARK MATRIX

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1. MICROBENCHMARKS

<i>INDEX</i>	<i>MICRO BENCHMARK</i>	<i>VARIED VARIABLE</i>	<i>SCHEMES RUN</i>	<i>METRIC</i>
1	Payload Size Scaling	Payload Size	• PSK • RSA + AES • CPABE + AES	• Encrypt Time • Decrypt Time • Memory • Ciphertext Overhead
2	Attribute & Key Scaling	• Attributes (CPABE) • Recipients (RSA) • Key Size (RSA)	• CP-ABE • RSA	• Encrypt Time • Decrypt Time • Ciphertext Size • Stored Key Size • Memory
3	Envelope Serialization	Attributes	• JSON • CBOR	• Serialized Envelope Size • Serialize Time • Deserialize Time • Memory
4	AES vs ASCON	Payload Size	• AES • ASCON	• Encrypt Time • Decrypt Time • Ciphertext Overhead

1.1. Payload Size Scaling

Measures how each full scheme's per-message cost grows as payload increases.

- **Encrypt Time:** Time to encrypt one message on publisher side
- **Decrypt Time:** Time to decrypt one message on subscriber side
- **Peak Memory**
 - **Peak RAM:** Highest RAM in use at any moment during the operation. Check against the device's RAM limit, if it exceeds, operation will crash
 - **Bytes allocated per op:** Total RAM grabbed during operation including temporary chunks released mid-way. Shows overall memory intensity of the operation.
- **Ciphertext Overhead:** Fixed bytes each scheme adds beyond raw payload (nonce, tag).

1.2. Expressiveness & Key-Size

Isolates asymmetric layer & measures how CP-ABE scales with policy attribute count, and on the other hand how RSA scales with recipient count, & key size.

- **Encrypt Time**
- **Decrypt Time**
- **Ciphertext Size:** For each schema grows in its own way
- **Stored Key Size:**
 - CP-ABE's secret key grows with the user's attributes
 - RSA's key bytes grow with modulus
- **Memory Allocation:** Amount of RAM in use during operations

1.3. Envelope Serialization

Compare packing exact same encrypted fields into a JSON envelope versus a CBOR envelope.

- **Serialized Envelope Size**
- **Serialize Time**
- **Deserialize Time**
- **Memory Allocation**

1.4. AES VS ASCON

Compares the two AEAD primitives in isolation over payload size

- **Encrypt Time**
- **Throughput:** Publisher & subscriber
- **Decrypt Time**
- **Tag + Nonce Overhead:** Bytes that cipher adds

2. MACROBENCHMARKS

INDEX	MACRO BENCHMARK	VARIED VARIABLE	SCHEMES RUN	METRIC
1	Sustained Load Profile	Message Publish Rate	• TLS-Only • PSK + AES • RSA + AES • CP-ABE + AES • CP-ABE + ASCON?	• E2E Latency • Actual Achieved Publish Rate • Actual Achieved Decrypt Rate • Publisher CPU & Memory • Subscriber CPU & Memory

2.1. SUSTAINED LOAD PROFILE

Steps the publish rate from a low baseline up to the point where the scheme visibly degrades, with either E2E latency growing unboundedly or the subscriber no longer keeping pace with publisher.

- **E2E Latency:** Delay from publish to plaintext delivery at the subscriber, capturing total pipeline cost
- **Actual Achieved Publish Rate:** Rate publisher actually sustains, which may fall below the target rate as the system saturates. Identifies the publisher side throughput ceiling.
- **Actual Achieved Decrypt Rate:** Rate at which subscriber completes decryption. Identifies the subscriber-side throughput ceiling.
- **Publisher CPU & Memory:** Tracks saturation on the publish side as rate increases.
- **Subscriber CPU & Memory:** Same on subscriber side.

NEW METRICS

METRIC	LEVEL	WHAT IT MEASURES
1	Energy per Operation	Micro & Macro
2	Thermal Behavior / Throttling	Macro
3	Peak Process Memory (RSS)	Macro
4	CPU Utilization	Macro

ENERGY PER OPERATION

Electrical energy consumed by a single cryptographic operation. Will be obtained by measuring the power draw while idle, then measuring power draw while it repeatedly performs operations. Energy per operation will then be derived by dividing total energy consumed over a time window by the number of operations completed in that window.

- Answers question "what does this scheme cost in battery life?"

THERMAL BEHAVIOR / THROTTLING

Measure CPU temperature of device over duration of run, and whether the device's thermal protection mechanism has activated.

- Help explain why throughput might degrade over time during a sustained load test, independent of any change in the code or workload.
- Logically applicable to macro scenario, where operation is continuous over a long enough window for heat to accumulate.

PEAK PROCESS MEMORY (RSS)

Measures maximum amount of RAM the running program occupies at any single point during a real workload.

- Answers the question "does this scheme's implementation fit within the memory budget of a constrained device?"
- Applicable to macro scenario since it is not meaningful at the microbenchmark level.

CPU UTILIZATION

Measures percentage of processor capacity in use while the device is under sustained load.

- Helps identify whether an observed throughput ceiling is caused by CPU or by some other bottleneck (e.g., network or I/O), which is necessary context for interpreting the throughput results in the Sustained Load Profile.
- Applicable to macro scenario since a single-threaded, non-blocking loop is CPU-bound by construction.